

22/12/22

Indian Institute of Engineering Science and Technology, Shibpur
 B.Tech CST 3rd Semester Final Examinations, December 2022
 Data Structures CS-2103

Full Marks: 50

Time: 3 hours

Attempt all questions

1. Answer all of the following questions

- a) Suppose a hash table T has 11 slots, $T[0], T[1], \dots, T[10]$ where all the slots are empty as shown below-

-	-	-	-	-	-	-	-	-	-	-
0	1	2	3	4	5	6	7	8	9	10

Also suppose 8 keys A, B, C, D, E, X, Y, and Z with the following hash addresses appeared

Keys:	A	B	C	D	E	X	Y	Z
$h(k)$:	3	7	1	10	3	10	4	0

Find the following with a brief explanation –

- Insert the keys into the hash table T as per the order shown above. If a collision occurs, follow the *open addressing* scheme to deal with it and state the final look of the hash table.
- Average number of probes(comparison) for a successful search (S)
- Average number of probes(comparison) for a unsuccessful search (U)
- The load factor (λ) of the hash table T.

b) Do the following for a Min-heap (ascending heap) with a brief explanation –

- Create a min-heap by inserting the following keys (26, 5, 77, 1, 61, 11, 59, 15, 48, 19) in that order. Draw the resulting min-heap after inserting each element.
- Draw the heap on performing delete operation (removing the root) twice.

$$(9+7 = 16)$$

2. Answer any 2 of the following questions using 'C' programming language

- a) Write the structure of an integer of size n, which can vary. Also, write createStack(...), push(...) and pop(...) functions.
- b) Assume a binary tree stores integer elements and is implemented following the linked-nodes approach. Write the binary tree node structure and a function to print all the external nodes (leaf nodes) of the binary tree.
- c) Given a singly linked list, write down a function that prints all the alternate nodes of it. For example, if the given linked list is $7 \rightarrow 3 \rightarrow 2 \rightarrow 4 \rightarrow 6$ then your function should print 7 2 6, and if the given linked list is $4 \rightarrow 2 \rightarrow 9 \rightarrow 4$ then it prints 4 9.

(5x2 = 10)

3. Answer any 6 of the following questions

- a) Write down the following
 - i. The postfix form of the infix expression: $D * (A + B) + E / (F + A * G) + C$
 - ii. The infix form of the postfix expression: $A B + C D * -$
- b) What is a binary search tree? What would be the worst case search time in a binary search tree having n nodes in it?
- c) Explain the process of deleting a node from a binary search tree.
- d) What is merging? Write down an algorithm to merge two sorted arrays to form a single sorted array.
- e) Explain the technique to perform insertion sort. From the implementation perspective, what kind of changes can you do to improve the performance of the Insertion sort?
- f) Assume an array L has the following elements in this order [13, 12, 9, 5, 12, 18, 7, 4, 21, 2, 6 11]. Consider the first element of the list (or partitioned list) as a pivot and illustrate step-by step operation of *PARTITION* on the array to achieve quicksort.
- g) Explain different methods for resolving collision in hashing and mention their advantages and disadvantages.
- h) Write a short note on "AVL Tree"

(4x6 = 24)